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Director of Waste Management
 c/o Liliana Jerade
 BC Ministry of Environment and Climate Change Strategy
 PO Box 9342 Stn Prov Govt
 Victoria, B.C. V8W 9M1

Attention: Liliana Jerade

Liliana.Jerade@gov.bc.ca

Re: Q3/Q4 2019 Progress Report for Management of High-Risk Conditions at the 5100 Portal Area of the former Sunro Mine, Jordan River BC, Site ID 16790

This letter provides a semi-annual update for the 5100 Portal Area of the former Sunro Mine ("Site"), as requested in the Ministry of Environment and Climate Change Strategy's (ENV) letter to Teck Resources Limited (Teck) and Western Forest Products Inc. (WFP), dated September 18, 2017.

In Q3 (August 30, 2019) Teck submitted the following reports to the ENV for Ministry review:

- Sunro Mine 2018 & 2019 Water Quality Monitoring Report, Sunro Mine, Jordan River, BC, dated August 28, 2019, prepared by SNC Lavalin; and
- Preliminary Aquatic Ecological Risk Assessment – Jordan River adjacent to the 5100 Portal Area of the former Sunro Mine, BC, dated August 2019, prepared by Azimuth Consulting Group Partnership.

Key findings from these reports were presented in the letter update to ENV sent on July 15, 2019.

In Q3 and Q4 2019, Teck also completed the final baseline water quality monitoring and sampling program; a preliminary evaluation and costing for soil management options; and an internal workshop to evaluate soil management options to support remediation and mine closure. The outcomes of these tasks are summarised below:

- The purpose of the sampling completed in Q3/Q4 2019 was to collect surface water, porewater and groundwater data representative of summer (dry) and fall (wet) conditions to characterize water quality to support remediation and closure planning under the Environmental Management Act¹ (EMA). The dry season is when the least amount of contaminant loading from the waste rock pile to the Jordan River was expected. Sampling was also completed in Dec 2019 to target the anticipated "worst-case" loading scenario of wet (rainy) conditions during low flows in the Jordan River where maximum flushing of the waste rock piles was expected with minimum dilution. The data will supplement previous investigations which delineated Protocol 12² high-risk conditions, established a conceptual site model and water balance to understand contaminant migration and loading pathways to the receiving waters of the Jordan River. The results of the water sampling program are currently being reviewed and interpreted.
- Closure scenario options identified in Q1/Q2 for waste rock were developed and costed in Q4. The identified options for waste rock at the Site included:
 - Option A: Stabilize waste rock *in situ* and minimize infiltration;
 - Option B: Move waste rock to new onsite landfill (subject to siting requirements, approvals, etc.);

¹ Environmental Management Act (EMA), B.C. Reg. 13/2019 / effective January 24, 2019.

² Protocol 12 for Contaminated Sites, Site Risk Classification and Reporting, BC Ministry of Environment & Climate Change Strategy, Version 2.0, BC ENV, March 12, 2013.

- Option C: Move waste rock to an existing permitted landfill remote from the site;
- Option D: Move waste rock to underground workings; and
- Option M: Partial removal of leachable mine wastes, stabilize remaining waste rock and minimize infiltration.

The designs were advanced from conceptual to pre-feasibility to support costing which included initial capital costs and net present value operating costs.

- The waste rock closure options were evaluated in an internal workshop using a multiple accounts analysis (MAA) process which evaluated each option against the project goals and remedial objectives, and considered the estimated costs and the interests of communities of interest (e.g., Western Forest Products, Pacheedaht First Nations, T'Sou-ke Nation, etc.). Although a preferred remedial plan was not selected at the workshop, it was determined that Option A (stabilize in-situ) and Option D (move waste rock underground) were not preferred and were dropped from the list of candidate closure plans. Options B (new on site landfill), Option C (off site permitted landfill) and Option M (partial removal/stabilization) are being considered further.

Work in progress or planned in Q1/Q2 in 2020, includes:

- Complete the report for 2019 water monitoring and sampling (Q1);
- Complete the multiple accounts analysis process and prepare a remedial and closure option report which presents the costs and preferred remedial plan for waste rock at the 5100 Portal Area of the Site (Q2);
- Consult with the communities of interest and regulators on the identified preferred remedial option (Q2); and
- Complete the water loading assessment to better understand flow dynamics within the mine and discussions with New Sunro Copper Group on water management to limit portal discharges (Q3/Q4).

The above activities will provide the necessary information to advance to the next step of remediation planning. As discussed, we anticipate consultation on a preferred plan in June 2020, with implementation of remediation to occur in 2021/22, subject to obtaining buy-in of the communities of interest, and permitting.

We trust this meets your current requirements. Please contact me if you have questions.

Yours sincerely,

Michelle Unger, B.Sc.
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cc. Steve Hilts, Teck Resources Limited
Christopher Dawes, Western Forest Products
Diane Howe, Ministry of Energy, Mines and Petroleum Resources